

Implement A*Algorithm

Aim :

To Find the shortest path from a source to a destination using the A* algorithm with heuristics.

Procedure:

1. Initialize a priority queue with the start node.
2. Maintain $g(n)$ (path cost) and $h(n)$ (heuristic).
3. Expand the lowest $f(n) = g(n) + h(n)$ node.
4. Track path using came_from dictionary.

Python Code:

```
from queue import PriorityQueue
def a_star(graph, start, goal, heuristic):
    open_set = PriorityQueue()
    open_set.put((0, start))

    came_from = {}

    g_score = {node: float('inf') for node in graph}
    g_score[start] = 0

    while not open_set.empty():
        _, current = open_set.get()

        if current == goal:
            path = []
            while current in came_from:
                path.append(current)
                current = came_from[current]
            path.append(start)
            return path[::-1]

        for neighbor, cost in graph[current]:
            tentative_g = g_score[current] + cost

            if tentative_g < g_score[neighbor]:
                came_from[neighbor] = current
                g_score[neighbor] = tentative_g
```

```

        f = tentative_g + heuristic[neighbor]
        open_set.put((f, neighbor))

    return None

graph = {
    'A': [('B', 1), ('C', 4)],
    'B': [('D', 2), ('E', 5)],
    'C': [('F', 1)],
    'D': [('G', 2)],
    'E': [('G', 1)],
    'F': [('G', 3)],
    'G': []
}

heuristic = {
    'A': 7,
    'B': 6,
    'C': 2,
    'D': 1,
    'E': 0,
    'F': 3,
    'G': 0
}

```

```
path = a_star(graph, 'A', 'G', heuristic)
```

```
print("Shortest path using A*:", path)
```

Output:

Shortest path using A*: ['A', 'B', 'D', 'G']

Result:

Thus, the A* Algorithm was implemented successfully.